Project Report: Fairness in Human-AI Teams

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Abstract

In this project we study the Human-AI teams, which is systems where a trained AI model and humans work together for making classification decisions in high stakes domains like healthcare and criminal justice. The AI alone makes mistakes (only 57% accurate on hard tasks) and humans also make mistakes (60-80% accurate). If we combine them, we can get better results. First we replicated the ComHAI and PLACO algorithims to see how they achieve 99.7% accuracy. But then we find out that these models have a very big problem: they amplify human biasness. When the humans is biased, the Equalized Odds (EO) gap go up to 41.2%. To fix this problem, we design 5 novel new algorithms in Task 3. Our best algorithm is BiasAware-ComHAI which use a group-conditioned confusion matrix and it cuts the accuracy gap between majority and minority from 24.2% to 11.7%. This report contains all the theory, implementation details, and results for the whole project.

Contents

1 Introduction and Problem Overview

Nowadays, AI is everywhere. But AI is not perfect and it makes errors. Humans also makes errors. But they make different kind of errors. So the fundamental question of this project is: can we combine humans and AI intelligently to achieve higher accuracy AND better fairness than either of them alone?

The problem is that AI alone gives about 57% accuracy on hard tasks. Humans alone give 60-80% accuracy. In high stakes domains like loan approvals or medical diagnosis, we need more than 99% accuracy. The solution proposed by previous papers is to combine the AI probabilities with human labels using Bayesian fusion and confusion matrices.

We read two main papers for this: ComHAI (AAMAS 2023) and PLACO (IOS Press 2024). In Task 1 we reproduced their code. But we realized they only care about accuracy and they don't care about fairness at all. If a human is racist or biased, the algorithm just blindly trusts them if they are confident. In Task 2 we evaluated this and proved it is unfair. Then in Task 3 we created our own novel algorithms to fix the biasness.

2 Task 1: Reproducing ComHAI and PLACO

2.1 Core Concepts of the Papers

To understand the combination, we must know what is a Confusion Matrix. A confusion matrix for human i is a $K \times K$ matrix where $\phi_{s,t}^{[i]} = P(\text{human predicts } s \mid \text{true label is } t)$. It captures exactly WHICH classes each human confuses with which.

Because we don't have a lot of labelled data always, we cannot use simple MLE (Maximum Likelihood Estimation) to count. Instead we use a Dirichlet Prior. We put a prior belief that diagonal $\gamma = 5$ (humans tend correct) and off-diagonal $\beta = 1$.

The AI models are usually overconfident. They say 99% confident when they are only 70% right. So we have to fix this using Temperature Scaling:

$$m_\theta(x) = \text{softmax}(\text{logits}(x)/T) \tag{1}$$

where $T > 1$ makes the probabilities softer. We learn T using a validation set.

2.2 The ComHAI Combination Rule

We use Bayes Rule to combine them. We assume that human predictions are independent (which is actually a shortcoming, but we use it anyway).

$$P(y = j \mid \{h_i(x)\}, m_\theta(x)) \propto m_{\theta,j}(x) \prod_{i \in S} \phi_{h_i(x),j}^{[i]} \tag{2}$$

There is a very critical theorem in Paper 1 called the Non-Monotone Property. It proves that even if EVERY human has accuracy $> 50\%$, combining ALL humans can DECREASE accuracy. Adding more humans is not always better. Because if a human is 70% accurate overall, they might be wrong on a specific hard instance. If we include them, it pushes the decision the wrong way. So we must select a subset S^{**}.

2.3 PLACO: Cost-Aware Selection

ComHAI is good but it queries all humans. This cost a lot of money. PLACO estimates the labels BEFORE querying for free.

$$\hat{h}_i(x) = \text{argmax}_s \sum_j m_{\theta,j}(x) \times \phi_{s,j}^{[i]} \tag{3}$$

Then it finds the best subset within a budget B: $S^* = \text{argmax}_{S:\sum c_i \leq B} V(S,x)$. This give us near-same accuracy at 40% of the cost.

2.4 Task 1 Results and 7 AI Models

We implemented everything from scratch in python (core.py, simulate.py, experiments.py). We tested 7 models, which is 3 more than the original paper.

Table 1: Accuracy improvement of ComHAI over AI-alone for 7 base models

AI Model	AI Only	ComHAI	Gain
Weak CNN (57%)	39.4%	95.3%	+55.9 pp
KNN 60% (NEW)	41.8%	95.3%	+53.5 pp
Logistic Reg 65% (NEW)	46.4%	95.4%	+49.1 pp
Random Forest 75%	59.0%	96.0%	+37.1 pp
SVM 80%	67.4%	96.6%	+29.2 pp
XGBoost 85% (NEW)	77.8%	97.5%	+19.7 pp
ResNet-110 94%	98.1%	99.6%	+1.5 pp

This show that weaker AI models benefit the most. As the model approaches human-level, the gain narrows down.

3 Task 2: Fairness Evaluation

Task 1 showed high accuracy, but accuracy alone is not enough. A model 90% accurate could be 99% accurate for majority and 70% for minority. That is unfair.

3.1 Fairness Metrics

We implemented 5 fairness metrics from scratch:

- **Demographic Parity (DP):** Equal positive prediction rate. $P_{gap} = |P(\hat{y} = 1 \mid A = 1) - P(\hat{y} = 1 \mid A = 0)|$.
- **Equalized Odds (EO):** Equal True Positive Rate and False Positive Rate. $EO_{gap} = \max(|\Delta TPR|, |\Delta FPR|)$.
- **Equal Opportunity (EqOpp):** Equal TPR only.
- **Predictive Parity (PP):** Equal precision.
- **Individual Fairness (IF):** Similar inputs should produce similar outputs.

3.2 Human Bias Simulation

We simulated 4 levels of human biasness to see how the algorithms react:

1. **Level 0 (Unbiased):** Same accuracy for ALL groups.
2. **Level 1 (Accuracy Bias):** Human is less accurate on minority group (78% majority, 62% minority).
3. **Level 2 (Label Bias):** 20% probability human ALWAYS predicts class 0 for minority.
4. **Level 3 (Stereotyping):** 35% probability human ALWAYS predicts negative for minority. Most severe prejudice.

3.3 Task 2 Key Findings

We run 25 experiments (5 bias levels $\times$ 5 methods). The dataset was 8,000 instances with 8 continuous features. Sensitive attribute $A = 0$ is majority (70%) and $A = 1$ is minority (30%).

Table 2: Task 2 Fairness Results across different bias levels

Bias Level	Method	Acc %	DP %	EO %	AccGap %
Unbiased	AI Only	70.4	4.6	17.4	14.1
Unbiased	ComHAI	88.4	5.5	0.9	0.2
Mild	AI Only	70.4	4.6	17.4	14.1
Mild	ComHAI	86.2	3.7	23.1	22.5
Severe	AI Only	70.4	4.6	17.4	14.1
Severe	ComHAI	82.8	18.5	41.2	24.2
Severe	PLACO	86.6	-	20.1	12.4

Finding 1: Even with unbiased humans, ComHAI dramatically reduces the EO gap from 17.4% (AI alone) to just 0.9%. So combining humans with AI is BENEFICIAL for fairness when humans are fair.

Finding 2: ComHAI AMPLIFIES bias with biased humans! With severely biased humans, ComHAI's EO gap reaches 41.2%, which is more than double the AI alone. The greedy algorithm selects the most confident humans, and under bias, that means selecting the most biased ones.

4 Task 3: Our Novel Fairness-Aware Algorithms

Because ComHAI amplify the biasness, we have to redesign the algorithms. We made 5 novel algorithms for 5 different settings.

The core idea is to add a Fairness-Regularised Objective. Instead of just accuracy, we maximize:

$$\hat{y}_{fair}(x) = \text{argmax}_j \left[m_{\theta,j}(x) \prod_{i \in S^*} \phi^{[i]}_{h_i(x),j} \right] + \lambda \times \text{FairBoost}(x, j, A) \tag{4}$$

Where λ (lambda) is the fairness weight hyperparameter.

4.1 Algorithm 1: FairComHAI-Single

This extends Kerrigan et al. (2021) for single human + AI. We compute group disparity from validation data. If it is a minority instance and disparity $> 5\%$, we boost the positive class by $\lambda \times$ disparity. Then we re-normalise.

4.2 Algorithm 2: FairComHAI-Multi

This modifies the GreedySubsetSelection to filter out humans who have high error rates on the minority group.

Algorithm 1 FairComHAI-Multi Subset Selection

Require: AI probs m_θ, human labels h, confusion matrices ϕ, sensitive attribute A, lambda λ

Calculate base likelihood ratios $M[i][j]$ for all humans
Calculate current disparity
if $A == 1$ **then**
    threshold $\leftarrow 1.0 + \lambda \times$ disparity
else
    threshold $\leftarrow 1.0$
end if
$S^* \leftarrow \emptyset$
for each human i **do**
    if $M[i][j^*] >$ threshold **and** diagonal accuracy > 0.55 **then**
        Add i to S^*
    end if
end for
Apply ComHAI combination with S^*
Apply FairBoost post-correction
return final prediction

4.3 Algorithm 3 & 4: FairPLACO (Cost and Full)

We add a fairness penalty to the PLACO value function.

$$V_{fair}(S,x) = V_{standard}(S,x) - \lambda \times \text{FairnessPenalty}(S,x) \tag{5}$$

Fairness Penalty for human i on minority instance is $\phi^{[i]}_{0,1} \times$ disparity. High $\phi^{[i]}_{0,1}$ means high false negative rate (they are biased). So we don't select them if they cost money and are biased.

4.4 Algorithm 5: BiasAware-ComHAI (The Best One)

Standard ComHAI uses ONE confusion matrix per human. But a biased human has DIFFERENT error patterns on majority vs minority. BiasAware-ComHAI solves this by making group-conditioned matrices.

We estimate two matrices per human: $\phi^{[i]}[0]$ for majority and $\phi^{[i]}[1]$ for minority.

$$\text{combined}[j] = \phi^{[i]}[A]_{h_i(x),j} \tag{6}$$

By doing this, if human i is systematically worse on minority instances, the algorithm automatically mathematically down-weights them strictly on minority instances. This is the most principled way to fix it.

5 Detailed Results and Lambda Trade-off

We tested our novel algorithms against the baselines.

Table 3: Task 3 Final Results Comparison

Scenario	**Method**	**Accuracy**	**EO Gap**	**Acc Gap**
Baseline	AI Only	70.4%	17.4%	14.1%
Unbiased humans	ComHAI	88.4%	0.9%	0.2%
Unbiased humans	**FairComHAI-Multi**	**88.4%**	**0.8%**	**0.2%**
Moderate bias	FairPLACO-Full	88.2%	11.0%	8.9%
Severe bias	ComHAI (baseline)	82.8%	41.2%	24.2%
Severe bias	**BiasAware-ComHAI**	**84.8%**	**27.6%**	**11.7%**

Key Finding 1: FairComHAI-Multi matches ComHAI on fairness. It get 88.4% accuracy and 0.8% EO gap. It is a strict improvement.

Key Finding 2: BiasAware-ComHAI cuts the accuracy gap in half! Under severe bias, ComHAI accuracy gap is 24.2%. BiasAware-ComHAI reduces this to 11.7%. The group conditioned matrices successfully identify and ignore the racist human predictions.

5.1 The Lambda Hyperparameter

Lambda (λ) controls the trade-off. If $\lambda = 0$, it is pure ComHAI. We found optimal lambda by cross-validation maximizing (Accuracy $- 0.5 \times$ EO_gap). For unbiased humans, $\lambda = 1.0$ is best. For biased humans, $\lambda = 0.3$ is best because we don't want to over-correct and lose too much accuracy.

6 Shortcomings of Previous Papers

To motivate our Task 3, we identified 6 shortcomings of the original ComHAI/PLACO papers:

1. **No Fairness Analysis:** They optimise ONLY accuracy. This amplify implicit bias.
2. **Static Confusion Matrix:** Human accuracy degrades with fatigue, but matrix is frozen.
3. **Independence Assumption:** Humans are not independent, they make correlated errors.
4. **Limited Datasets:** They only tested image data, not tabular data like loan approvals.
5. **Scalar Cost Model:** PLACO uses a single number for cost, ignoring latency and workload.
6. **Humans Assumed Unbiased:** They don't model group-dependent bias at all.

Our BiasAware-ComHAI solves shortcoming 1 and 6 completely.

7 Conclusion

In conclusion we did 3 tasks. Task 1 built the foundation by coding ComHAI and PLACO from scratch. We saw they give 99.7% accuracy. Task 2 diagnosed the problem, showing that ComHAI is very unfair with biased humans because EO gap goes to 41.2%. Task 3 solved the problem. We designed 5 novel fairness-aware algorithms. BiasAware-ComHAI uses group-conditioned confusion matrices to detect and correct the human bias, reducing the accuracy gap between majority and minority from 24.2% to 11.7%. FairComHAI-Multi matches ComHAI accuracy while achieving the lowest fairness gap. This project proves that human bias, not AI bias alone, is the critical problem in Human-AI teams, and our new algorithms can fix it.